PROBLEM SET 1, PART 1: SMOOTH MANIFOLDS
DUE: SEP. 26

(1) [**Topological group**]
A topological group is a topological space G which is also a group, so that the multiplication operation

$$\mu : G \times G \to G, \quad (g_1, g_2) \mapsto g_1 g_2$$

and the inverse operation

$$i : G \to G, \quad g \mapsto g^{-1}$$

are both continuous. For any subsets S and T in G we can define

$$ST = \{g_1 g_2 \mid g_1 \in S, g_2 \in T\}.$$

By this way we can define the subsets $S^n = S \times S \times \cdots \times S$ and S^{-1}.

(a) If G is a topological group, and U is any open neighborhood of the identity element $e \in G$. Prove: There exists an open neighborhood V of e so that $V = V^{-1}$ and $V^2 \subset U$.

(b) Prove: If G is a connected topological group, then for any open neighborhood U of the identity element $e \in G$, we have

$$G = \bigcup_{n=1}^{\infty} U^n.$$

(c) [Not required] Suppose G is compact Hausdorff topological group and $g \in G$. Prove: $e \in \overline{\{g^n \mid n \in \mathbb{Z} \setminus \{0\}\}}$.

(2) [**Locally Euclidean**]
Prove the following properties of locally Euclidean spaces:

(a) Any connected component of a locally Euclidean space is open.

(b) Any connected locally Euclidean space is path connected.

(c) [Not required] Any locally Euclidean Hausdorff space is regular. [Thus as a consequence of Urysohn's metrization theorem, any topological manifold is metrizable.]

(d) [Not required] If both X, Y are connected, second countable and locally Euclidean, and $f : X \to Y$ is bijective and continuous, then f is a homeomorphism.

(3) [**Topological manifolds with boundary**]

(a) Find the definition of topological manifolds with boundary from literature.

(b) Prove: If M is a topological n-manifold with boundary, then its boundary, ∂M, is a topological $(n-1)$-manifold without boundary.

(c) [Not required] Prove: the product of two topological manifolds with boundaries is a topological manifold with boundary. What is its boundary?

(4) [**Connected topological manifolds are homogeneous**]
Let M be a connected topological manifold. Prove: for any $p, q \in M$, there exists a homeomorphism $\varphi : M \to M$ so that $\varphi(p) = q$.

(5) [**Local homeomorphism**]
Let X, Y be topological spaces. A map $f : X \to Y$ is called a *local homeomorphism* if for every point $x \in X$, there exists an open set U containing x such that the image $f(U)$ is open in Y, and the restriction $f|_U : U \to f(U)$ is a homeomorphism (with respect to the respective subspace topologies).
   (a) Show that every local homeomorphism is an open map (i.e. maps each open set to an open set).
   (b) Show that if a local homeomorphism is bijective, then it is a homeomorphism.
   (c) Show that if Y is locally Euclidean and $f : X \to Y$ is a local homeomorphism, then X is locally Euclidean.
   (d) Show that if X is locally Euclidean and $f : X \to Y$ is a surjective local homeomorphism, then Y is locally Euclidean.